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Chapter 1 — Repository Inspection Report

This chapter reports the state of the connected repository (github.com/jamesrudh-arch/Jarvis) exactly as found at Phase Zero revision 2. The first inspection found an empty scaffold; a merged pull request has since delivered a substantial implementation claiming Milestones 1-4. This revision inspects that implementation by direct code reading. The inspection remained read-only: no files were deleted, overwritten or reinitialised, no commits or pushes were made, no application source was modified and no dependencies were installed. The only artifacts created during Phase Zero are the pre-production documents and storyboard assets delivered alongside this package.

1.1 What the repository contains

Branch **main**, three commits: the initial scaffold, an implementation commit **d3a25f9** ("Implement JARVIS NEXUS Milestones 1-4") arriving via merged pull request #1 from branch **arena/3bdc7997-jarvis**, and the merge commit itself. 62 tracked files across four modules: a Python FastAPI backend (about 3,800 lines - 47 REST routes, 2 WebSocket endpoints, 16 SQLAlchemy models, services, an SQL migration, 7 automated tests), a Vite + Three.js frontend (about 5,200 lines - sanctuary scene, character controller, camera director, scenario director, procedural textures, MIT-licensed vendor modules for dynamic bones, lip-sync and retargeting), a private inference worker (290 lines), and a docs/preproduction folder with its own baseline document and 18 storyboard plates. Model assets: a 13.8 MB rigged 66-bone GLB character with 72 ARKit/Oculus morph targets, plus Mixamo locomotion clips.

1.2 What is already functional (verified by code reading)

Substantially more than the first inspection found, and the claims largely check out against the code: server-side enforcement of all seven owner switches (voice input returns 403 with the microphone switch off); a ten-category epistemic memory system with Fernet encryption of sensitive records and a hard HTTP 409 guardrail preventing silent promotion of hypothesis to verified fact; goal dependency scheduling with audit events; bounded specialist agents with dispatch and cancellation; allowlisted tools with audit logs; encrypted Kaggle profile management with genuine package export; an authenticated worker WebSocket; and - importantly - a **truthful offline executive**: when no GPU worker is connected, conversation falls back to deterministic replies that explicitly label themselves as coming from an offline source rather than fabricating model output. The frontend implements the twelve zones and fifteen modes, Mixamo retargeting with root-drift stripping, gaze-first turning, blinking, phoneme-to-viseme lip-sync, and a scenario director with voxel clouds, variables, timeline scrubbing and world restoration.

1.3 What can be retained

Essentially the whole implementation skeleton. The architecture matches the product specification: FastAPI + SQLAlchemy with SQLite for development and PostgreSQL + pgvector for production behind DATABASE_URL, WebSocket-first live state on /ws/nexus and /ws/worker, server-side secret encryption, and a worker design using the default model huihui-ai/Qwen2.5-14B-Instruct-abliterated with hardware-aware 4-bit NF4 quantisation, Kaggle Secrets authentication and zero public ports. The twelve-zone world layout matches Chapter 3 of this package zone for zone. The seven switches are enforced on both sides. Nothing needs to be thrown away; the work forward is verification, visual alignment and hardening rather than replacement.

1.4 What is missing or must change

Visual identity. The repository's character baseline predates the owner's approved direction. Its storyboard plates and character reference were generated without identity locking - independent QC confirms re-rolled faces between plates, waxy skin and vacant eyes - and its baseline document self-describes as "Approved by Owner", which is not accurate: the owner's actual direction is the identity anchor locked in Chapter 2 of this package. The shipped 3D character's wardrobe and appearance must be aligned to that anchor (gate M0.5 in Chapter 6). The repo's own storyboard plates are superseded by the twenty identity-locked frames in Chapter 4.

Runtime verification. The automated tests were authored by the implementing agent and have not been executed in the owner's environment; the verification protocol in Section 6.3 applies in full, including a real browser session, real locomotion inspection and a real GPU worker conversation before any capability is called verified. **Production hardening** (deployment configuration, CI, backup and restore drills) and **license hygiene** (retargeter.mjs carries no license header; Avaturn GLB provenance to be confirmed in writing; Mixamo clips fall under Adobe terms) complete the list.

1.5 Existing architectural and design decisions

The implementation encodes decisions this package adopts as the baseline: a monorepo (backend / frontend / worker / docs); browser-first Three.js rendering with ACES filmic tone mapping and restrained bloom; the twelve-zone, fifteen-mode interaction model inside one continuous world; seven hardware-grade owner switches as the authority model; typed memory with epistemic statuses; goal dependency gates; and private-first inference (local PC or Kaggle GPU, no mandatory cloud API, Ollama and OpenAI-compatible endpoints as optional adapters only). The earlier hygiene defect from the first inspection is resolved: .env is no longer tracked and .env.example is committed in its place.

1.6 Technical, licensing and asset constraints

Third-party code carries obligations that are accounted for: dynamicbones.mjs and lipsync-en.mjs are MIT-licensed (Mika Suominen, 2024) and retained; retargeter.mjs appears original but carries no license header and must gain one at M0.5; the character GLB is Avaturn-generated and its commercial-use terms must be confirmed in writing before ship; Mixamo animation clips fall under Adobe's terms for interactive use. Constraints inherited from the product brief remain binding on all milestones: no Ollama or commercial AI API may be required; MetaHuman is optional, never a prerequisite; holographic output must be a live view of the running application; inference, metrics, agent work and test results are never fabricated. No repository change was made during Phase Zero, per the inspection rules.

Module	Content	Assessment
backend/app	FastAPI core: 47 REST routes, 2 WS endpoints, 16 models, inference hub, world hub, simulation and tools, seed data, Fernet crypto	Retain. Claims verified by code reading; runtime verification pending (Section 6.3)
backend/migrations, tests	Initial pgvector-ready SQL schema; 7 test functions covering switches, memory guardrails, goals, scenarios, worker streaming	Retain; extend with owner-environment runs before any capability is called verified
frontend/src/world	NexusSanctuaryScene (12 zones), JarvisCharacterController (66-bone rig, 72 morph targets, gaze, blink, lip-sync), CameraDirector, ScenarioDirector3D, ProceduralTextures	Retain as the renderer base; visual identity of the character aligns to the approved anchor at M0.5
frontend/src/vendor + models	MIT dynamic bones and lip-sync modules; retargeter; jarvis_avaturn.glb (13.8 MB); Mixamo walking and locomotion sets	Retain subject to license hygiene: header on retargeter, written Avaturn terms confirmation
worker/jarvis_worker.py	Private model engine: Qwen2.5-14B default, NF4 4-bit CUDA quantisation, Kaggle Secrets, outbound authenticated WS, streaming, cancellation	Retain; verify with a real GPU session before calling inference verified
docs/preproduction	Repo's own baseline document + 18 storyboard plates + 2 asset references	Superseded: plates are not identity-locked (QC-confirmed face re-rolls); replaced by this package upon approval
scripts, .env.example, README	Start scripts, worker runner, environment template, honest capability notes	Retain; .env no longer tracked - first-inspection hygiene defect resolved

Table 1.1 - Repository modules and assessment (62 tracked files, 3 commits)

3 commits · 62 tracked files · 47 REST routes + 2 WebSocket endpoints · approx. 9,300 LOC

Repository state at Phase Zero revision 2 - an M1-M4 skeleton implemented, pending owner-environment verification and identity alignment

Chapter 2 — Character Direction

JARVIS is an adult female intelligence with a poised, perceptive, professional presence. Revision 2 locks her appearance to the owner-approved identity anchor (Plate 2.1): every concept plate and every storyboard frame in this package is generated as a controlled edit of that single reference image, so one woman - the same face, hair, wardrobe and watch - carries across all twenty frames; independent QC scored identity continuity at 8-9/10 frame-to-frame. The direction remains buildable: every visual trait maps to the licensed asset pipeline in Section 2.9 and the animation systems in Chapter 5. The shipped 3D character comes from the existing repository rig aligned to this identity, not from these images.

2.1 Age presentation and physical realism

JARVIS presents as a woman of about thirty-seven. Maturity is carried by structure, not decoration: a defined jaw line, mature brow-to-chin proportions, natural cheek planes, and skin that behaves like skin - three-layer subsurface scattering, visible pores, faint under-eye shading and subtle asymmetric imperfections. She is never neotenous, doll-like or plastic. Her body carries believable weight: mass transfers through the hips on turns, strides load the knees, and cloth and hair react to real velocity. Resting expression is calm attentiveness - emphatically not a smile.

2.2 Face and hair

The face follows the approved anchor: a slim oval with high cheekbones and an angular, defined jaw; straight, slightly full dark brows; grey-brown eyes with true corneal specular; medium-full lips held closed and neutral at rest. Hair is dark brown, secured in a low bun that keeps the face fully visible, with a few loose strands for realism. Hair is shaded anisotropically and moves with physically plausible inertia - no billboard sheets, no helmet shine, and never coloured or flowing fantasy hair.

2.3 Wardrobe

The baseline is the **Nexus duty suit**, locked to the anchor: a black fine-knit turtleneck under a tailored structured dark charcoal blazer, slim dark trousers, low practical boots, and a wristwatch worn on the left wrist. The silhouette is serious professional operations wear - precisely what the approved reference shows. There is no armour, no catsuit, no glowing costume lines, no lingerie styling, no science-fiction costume cliches: her wardrobe would not look out of place in a serious operations centre, which is exactly the point. A formal variant may add an unstructured drape coat for presentation moments (decision D3).

2.4 Materials and colour treatment

Materials are separated by family and response: skin (subsurface, soft key response), hair (anisotropic sheen), wool-knit and technical twill (matte, weave-level micro normal), leather touchpoints, and worn metal hardware on clasps and the earpiece. Her wardrobe never emits light. Cyan exists on her only as reflection from consoles and displays; amber alert washes reach her the same way they reach the architecture. The world lights her - she is not a light source.

2.5 Body proportions

Realistic adult female proportions at approximately 174 cm: natural shoulder-to-hip ratio, anatomical joint placement, and corrective blendshapes at shoulders, armpits and hips so deformations stay clean in lean, reach and console poses. No exaggerated anatomy of any kind. The rig is complete - full skeletal hierarchy, hand and finger articulation, foot roll bones, and an ARKit-52 facial blendshape set.

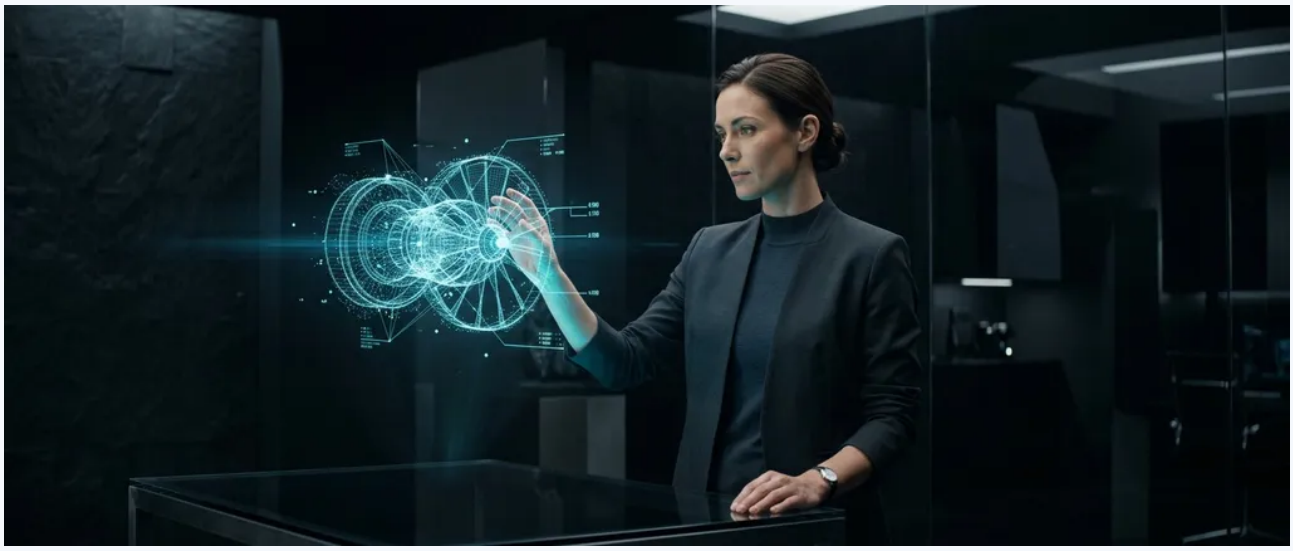


Plate 2.1 - The owner-approved identity anchor. This single reference image is the law of the project: every concept plate and storyboard frame in this package is a controlled edit of it.





Plates 2.2-2.3 - The locked full-length character sheet (left) and face study (right), both generated as controlled edits of the anchor - one woman, guaranteed. The shipped 3D character comes from the repository rig aligned to this identity (Table 2.3).

2.6 Expressions

Five expression states cover the working range. Each is defined in FACS action units so the facial rig, the animation system and any performance capture pass all share one vocabulary, and each is a modulation of a living baseline - never a frozen mask.

Expression	Facial action (FACS)	Body and timing cues
Neutral	Baseline; lids relaxed, jaw unengaged	Attentive rest; weight 60/40; breathing 4 s cycle; blink 2-5 s
Listening	AU1+2 trace (brow raise), head tilt 3-5 degrees	Still posture; sustained gaze; slow blink cadence; micro AU12 on comprehension
Focused	AU4 (brow lower) + AU5 trace (lid tension), lips closed	Lean toward task; blink interval 3-6 s; micro jaw set; gestures slow
Concerned	AU1+4 (worry combination) + AU15 trace (corner depress)	Gaze shifts to evidence; breath shallower; hands quiet or one brace
Urgent	AU5+7 (wide, tighten) + AU23 (lip tighten), AU26 on speech	Forward lean; weight-shift tempo up; stride length up; no panic, no flail

Table 2.1 - Expression set (FACS-coded). Restraint is the rule; each state is a modulation, never a mask.

2.7 Poses and locomotion targets

The five posed states below are the building blocks every mode composes from; speed and transition targets are contractual values the controller is tuned against.

State	Direction
Standing (attentive rest)	Weight 60/40 on one leg, arms relaxed, clavicle level, gaze anchored on the primary stimulus; ready, not rigid
Walking	Purposeful 1.15-1.25 m/s; arm swing from the shoulder; tall spine; steady head; no model glide
Running	Measured urgency 3.2-3.8 m/s; forward lean 8-12 degrees; stride extension; jacket and hair respond
Console interaction	Hinged 15-20 degrees at the hips; one hand braced or hovering; the other tracks interface targets; feet planted
Presentation	Open stance 30-45 degrees to the display; free hand gestures toward the media; she stands beside content, never in front of it

Table 2.2 - Core poses and locomotion targets

2.8 Facial animation and lip-sync approach

The facial stack is layered, lowest to highest: visemes, expression, blink, gaze. Speech drives an ARKit-52 viseme set (9-12 visemes with co-articulation) from phoneme timings supplied by the TTS adapter - timing comes from the audio pipeline, never guessed from text. The expression layer composites above visemes at roughly 60/40 weight so speech stays legible while emotion stays live. The blink system runs a 2-5 s Poisson cadence (120 ms close, 200 ms open) with occasional thought-blinks linked to turn-taking. Gaze uses saccades, smooth pursuit and micro-saccades; the head leads the eyes beyond 20 degrees of offset. Prosody stress points earn at most two or three emphasis beats per sentence - restraint is a hard budget, not a mood. In silence the face drifts back to neutral; it never freezes.

2.9 Asset pipeline

The pipeline below is repository-first. Every entry names its licence basis and role; nothing enters a release without that row filled in.

Layer	Source	Licence basis	Role
Character (primary)	Existing repository rig: jarvis_avaturn.glb (Avaturn-generated, 66 bones, 72 ARKit + Oculus morph targets)	Avaturn terms - commercial use to be confirmed in writing at M0.5	Base rig; wardrobe and appearance aligned to the approved anchor via garment/material update or regeneration
Character (M4 optional)	Epic MetaHuman	MetaHuman EULA (free)	Optional higher-fidelity body for the existing NPC - never a prerequisite
Locomotion	Existing Mixamo clips (walking.fbx, xbot_locomotion.glb) + ActorCore extensions as needed	Adobe terms (interactive use) + Reallusion content licence	Blend-space clips with authored transitions; root-drift stripping already implemented
Gestures and interaction	Mixamo library + bespoke keyframe	Adobe Terms + project-owned IP	Gestures, console and presentation sets
Lip-sync and hair dynamics	Existing MIT vendor modules (lipsync-en, dynamicbones)	MIT (Mika Suominen)	Phoneme-to-viseme scheduling and secondary hair motion, already integrated
Viseme pre-pass (optional)	NVIDIA Audio2Face or equivalent adapter	NVIDIA terms (free for use)	Replaceable adapter feeding the ARKit viseme set

Table 2.3 - Asset pipeline (repository-first; nothing unlicensed ships)

Selection is gated on a Milestone 1 asset spike: a one-week evaluation in which the candidate character is imported, retargeted and driven through the full locomotion and facial stack before any licence is purchased in bulk. If no licensed asset meets the quality bar, the limitation will be stated precisely and a custom commission scoped - never concealed with a cheap substitute.

Chapter 3 — Environment Direction

The world is one continuous interior - **The Nexus** - a subterranean operations floor of roughly 30 by 18 metres with 4.2 metres clear height and a mezzanine over the east wing. The architectural language is monolithic and quiet: honed near-black basalt volumes, dark brushed steel, machined reveals carrying hairline cyan light, an acoustic ceiling with linear apertures, and a raised floor with inset guide lighting. It is not a neon cave, not a spaceship corridor, not a showroom: it is a working facility that happens to be beautiful the way serious infrastructure is beautiful.

3.1 The floor plan

Twelve areas open onto one continuous floor. The plan below is the spatial contract: every mode, camera framing and scene in this package is composed against it.

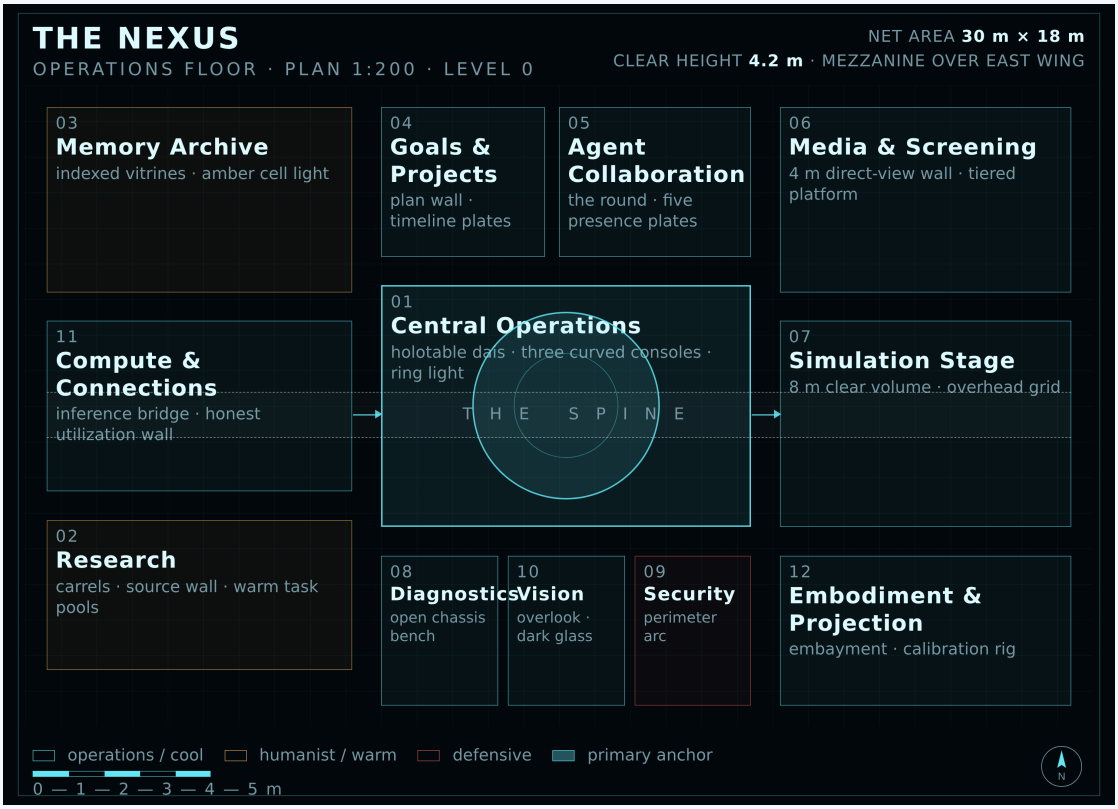


Plate 3.1 - The Nexus, operations floor plan. Cool zones carry work; warm zones carry memory and study; the defensive arc alone holds red.

No.	Area	Spatial identity and equipment	Key interaction points
01	Central Operations (Hub)	Circular dais with a low direct-view holotable, three curved consoles, overhead ring light; the compositional anchor of the floor	Task summoning, world overview, status glance, owner console at the hub edge
02	Research (Library)	Study carrels with dual curved displays and a physical source wall; warm 3000 K task pools against the cool ambient - a humanist study inside the facility	Source pinning, citation tracing, evidence extraction, thread steering

No.	Area	Spatial identity and equipment	Key interaction points
03	Memory Archive (Vault)	Glass-fronted indexed vitrines with warm amber cell light; deeper column rhythm, slower pacing - memory reads as warm and kept	Recall, pin, correct, forget; provenance chips separate fact, preference and inference
04	Goals and Projects (Plan Wall)	Longitudinal wall of persistent project plates with light timelines and dependency lines; standing bench	Approve, reweight, pause, resume; auditable next-action proposals
05	Agent Collaboration (The Round)	Semicircular table with five recessed presence plates - abstract light presence, deliberately not humanoid robots; JARVIS stands at the open arc	Listen-in, promote or veto agent actions, request transcripts
06	Media and Screening	Four-metre direct-view wall, two-step tiered platform, fully dimmable zone	Play, pause, seek, frame-step, A-B compare, frame extraction, 2D-to-3D handoff
07	Simulation Stage	Eight-metre clear volume with overhead grid, reconfigurable floor projection and boundary markers; may borrow adjacent floor for scale scenarios	Scenario construction, playback, variables, branching, actor direction
08	Diagnostics	Open-chassis bench, oscilloscope culture, honest cabling, live telemetry displays	Health inspection, trace capture, fault isolation
09	Defensive Security	Quieter inward-facing arc of monitors; red is reserved for confirmed-critical states only	Anomaly review, evidence chain, confirmation-gated containment
10	Vision (Overlook)	Dark glass wall; live external views render only when the camera switch is genuinely on	Explicit-activation viewing, capture-state truth, scene understanding
11	Compute and Connections (Bridge)	Restrained compute wall with honest utilisation lighting - no infinite server racks; the bridge desk where inference profiles physically dock	Local/Kaggle profile management, genuine auth tests, worker packages
12	Embodiment and Projection (Embayment)	Raised platform with calibration rig and capture volume; the projection surface draws from the same live instance as every other view	Hologram layout control (mirror, rotate, multi-face), live sync

Table 3.1 - The twelve areas (numbers match the floor plan)

3.2 Lighting, materials and navigation

Lighting is motivated: fixtures are visible, pools are purposeful, and colour is information. Warm 2700-3000 K pools mark human zones (Library, Vault, Plan Wall); cool 4000-5000 K marks machine zones (Hub consoles, Bridge, Diagnostics). Cyan is reserved for information - displays, guide lines, status - and never washes whole surfaces; amber marks attention; red appears only on confirmed-critical states. JARVIS is lit with discipline: a soft 3/4 key that keeps her face readable everywhere she works, a cyan rim borrowed from nearby consoles, and warm bounce when she stands in humanist zones. Volumetric depth comes faintly from ceiling apertures only - no fog banks hiding weak assets. Contact shadows are always on; bloom is restrained.

Material families stay distinct: honed basalt, brushed dark steel, smoked low-iron glass, muted blue-grey acoustic felt, leather at touchpoints, worn metal hardware, and emissive displays. Architecture does not glow; information does. Navigation runs along **The Spine**, the east-west axis every zone opens onto, with breathing guide lines and a secondary ring behind the Hub. Wayfinding is task-linked: when JARVIS works, she goes to where the work lives, and the floor plan below is the contract every mode, camera and scene obeys.

Swatch	Colour	Role in the world
#02070B	Near-black blue	Base darkness of architecture and shadow; the floor's resting tone
#06121A	Architectural surfaces	Basalt and steel panel bodies; large-area structure
#62E6F2	Primary cyan	Information only - displays, guide lines, status, annotations; never ambient wash
#DDFBFF	Ice white	Text on dark surfaces, timeline light, highest-emphasis highlights
#7799A5	Muted text	Secondary labels, inactive states, captions inside the world
#E7AE5B	Warning amber	Attention states, alerts-in-progress, sync progress, blocked dependencies
#FF6B67	Critical red	Confirmed-critical only - used sparingly, never decoratively

Table 3.2 - In-world palette (the world's own brand, used by renderer, UI and scenario systems)

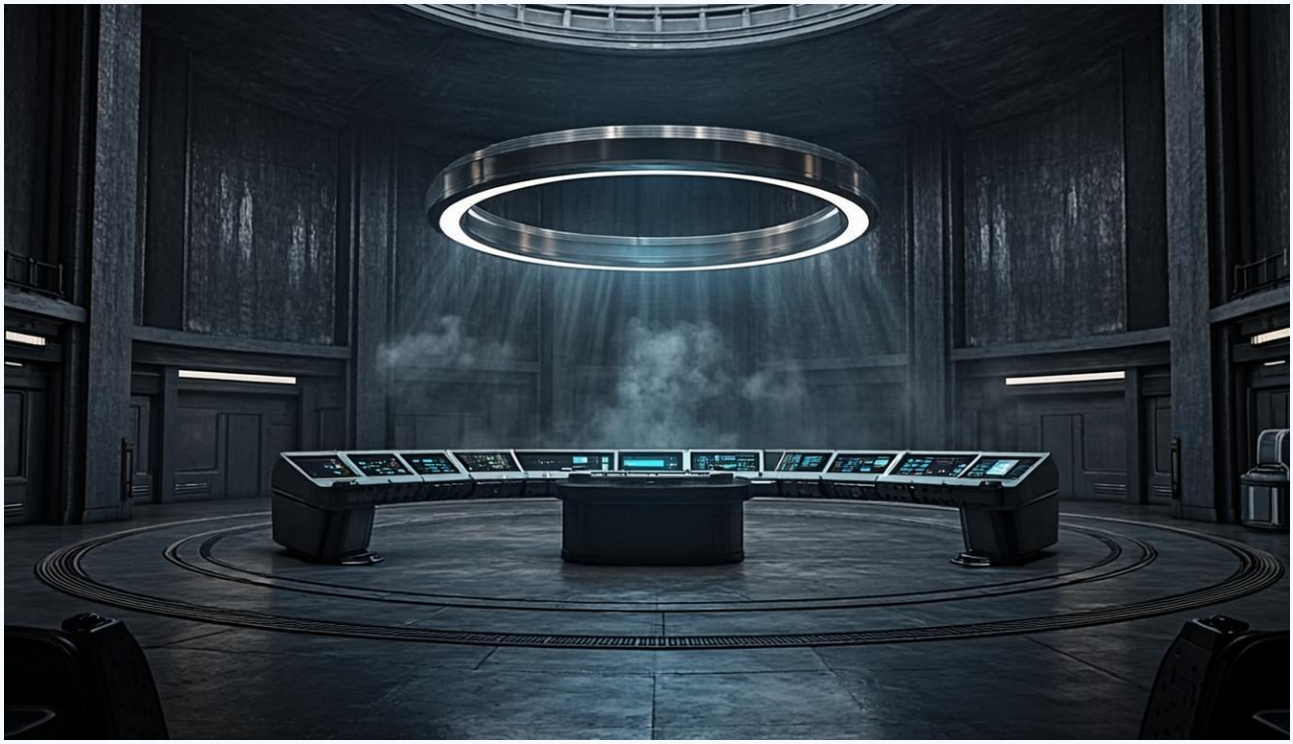


Plate 3.2 - Central Operations: dais, holotable and console ring under the overhead light ring.

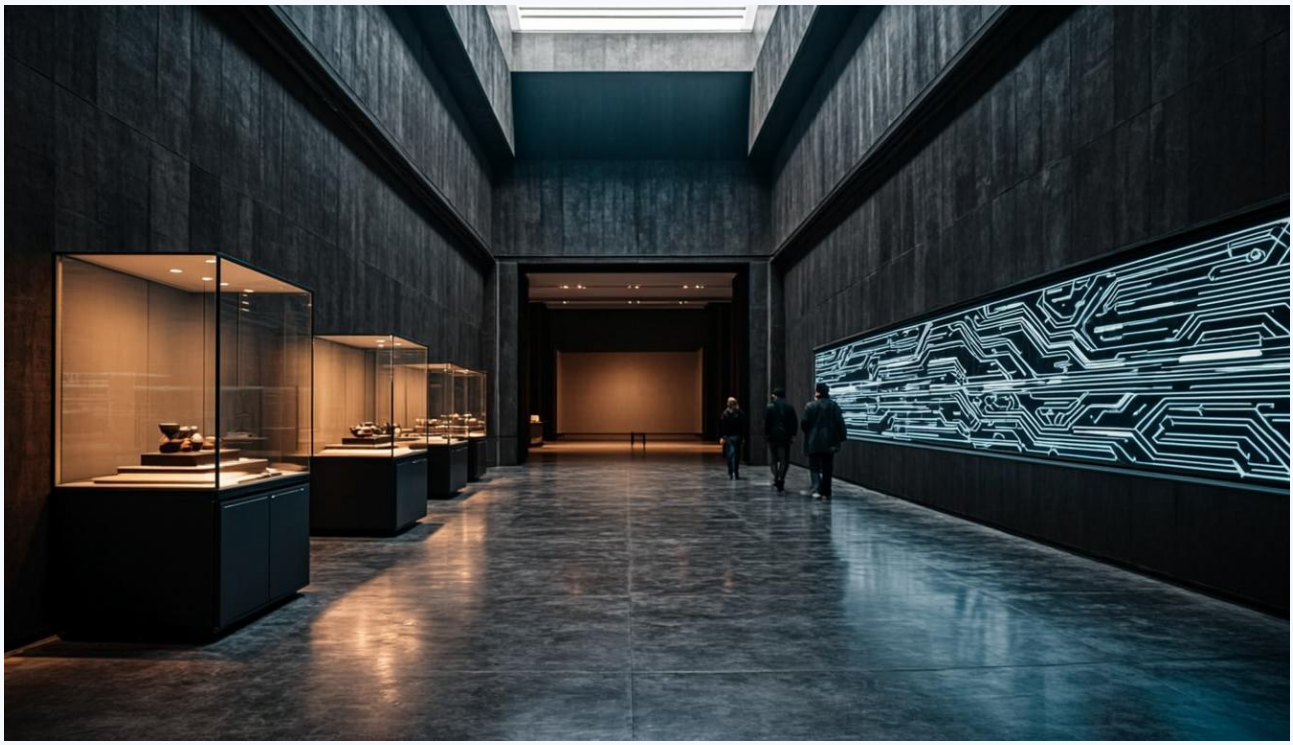


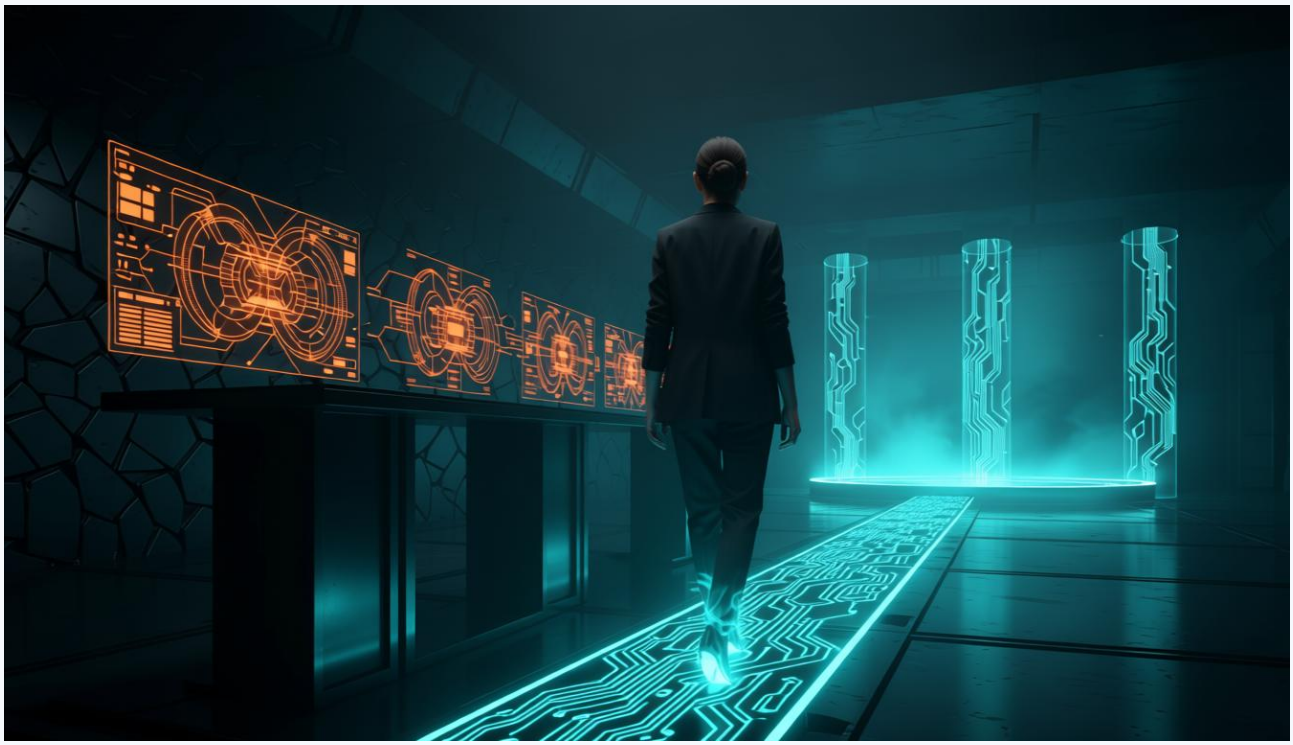
Plate 3.3 - The Spine axis: Memory Archive warm-left, Plan Wall right, the Stage volume in the far distance.

Chapter 4 — Visual Storyboard

Twenty frames, one per page. Each frame is an individually composed concept study that fixes action, emotion, camera, lighting, environment, visible information, owner interaction and transitions. The images define composition and intent; the shipped character and world come from the licensed pipelines in Chapter 2 and the architecture in Chapter 3. Frames are numbered to match the mapping matrix in Chapter 5, and each carries the operational mode it serves.

Frame 01 · FIRST WAKE

MODE 1 · WAKE & AUTHENTICATION — THE SPINE → HUB



Action and task state	Cold-start sequence completes; consoles cascade online along the Spine. JARVIS walks in from the west; her first anchor is the hub holotable, then the owner console.
Emotional state	Composed emergence - neutral baseline, gaze settling. No theatrical entrance, no speech yet.
Camera and framing	Wide establishing, 24 mm, low angle from the Spine floor. JARVIS at the right third, small-to-medium in frame; 1.5 s exposure ramp from black.
Lighting	Cold-start state: consoles cascading, single hub ring overhead, her figure rim-lit cyan at low intensity against deep ambient shadow.
Environment	The Spine and Hub; floor guide lines breathing; Memory Archive dark to the west.
Visible information	Boot truth only: CORE online · memory intact · inference: none. The owner console glows, awaiting authentication.
Owner interaction	None yet - authentication prompt on the owner console; free camera observation is available.
Entry / exit transitions	IN: from black, slow exposure ramp. OUT: she turns toward the owner console; camera begins a slow push, cutting on turn completion.

`SYSTEMS · renderer cold-start pass · CORE boot truth · owner console auth · camera exposure ramp`

Frame 02 · NORMAL LIVING PRESENCE

MODE 2 · LIVING PRESENCE — CENTRAL HOLO-RING



Action and task state	Unaddressed background work at the holo-ring: she assembles a turbine structure element by element above the pedestal table, one part half-attached and trailing voxel particles. She glances up as the owner's presence engages.
Emotional state	Quiet focus breaking into attentiveness; mild recognition (brow-raise trace). No broad smile, no greeting performance.
Camera and framing	50 mm medium-long from the hub axis; body three-quarters to the table; the head turn completes as the camera settles.
Lighting	The hologram is the practical key - cyan on her face and hands; ring glow from the pedestal; deep ambient shadow around.
Environment	Central holo-ring, pedestal table; one glass data column glowing faintly in the background.
Visible information	Three or four project plates with statuses (in progress / scheduled / blocked), a next-action line, one amber dependency.
Owner interaction	Passive. She acknowledges with a brief glance and a single line; the owner may open conversation.
Entry / exit transitions	IN: cut. OUT: dolly-with as she steps away toward the Hub on a queued task.

`SYSTEMS · ambient task scheduler · plan-wall surface · presence glance system`

Frame 03 · WALKING TOWARD CENTRAL OPERATIONS

MODE 2 · LIVING PRESENCE — THE SPINE



Action and task state	Purposeful walk at 1.2 m/s along the Spine toward the Hub; earpiece dot amber (task queued); arm swing natural, stride matched to distance.
Emotional state	Neutral purpose; baseline face. Moving because there is a reason, never pacing for effect.
Camera and framing	Tracking dolly, 50 mm, trailing three-quarter rear at 2.5 m with a slight low bias; stabilised, no drift.
Lighting	Guide lines beneath, intermittent ceiling apertures casting soft pools; passing consoles as cool rim sources.
Environment	The Spine, west to east.
Visible information	Floor guide line to the Hub; one task chip (compiling morning digest). Nothing else competes for attention.
Owner interaction	The owner may hail at any time; she transitions into a stop-and-plant-turn.
Entry / exit transitions	IN: from Frame 02 follow-on or cut. OUT: match-move into Hub arrival as the camera re-frames on deceleration.

`SYSTEMS · locomotion blend space · navmesh spine route · follow-camera dolly`

Frame 04 · TURNING TO ACKNOWLEDGE ME

MODE 3 · CONVERSATIONAL FOCUS — HUB EDGE

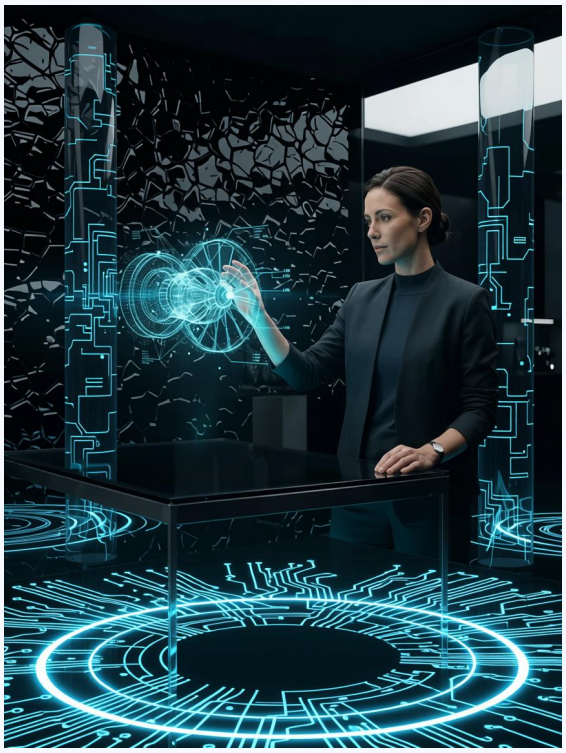


Action and task state	Addressed mid-stride: deceleration, plant-turn of roughly 120 degrees in place. Head leads, eyes lock the owner camera, weight settles.
Emotional state	Attentive recognition; a micro brow-raise; mouth neutral and ready to speak.
Camera and framing	Owner point of view, 40 mm at eye height, medium-close. Her turn ends exactly on the camera axis.
Lighting	Hub ring key overhead, soft; cyan rim from the console behind her; face fully readable.
Environment	Hub edge; consoles idle-cool around her.
Visible information	Deliberately minimal - one status chip dissolves; her attention is the information.
Owner interaction	Full. She is listening; voice and text input are live; she remains interruptible throughout.
Entry / exit transitions	IN: cut on the hail. OUT: settles into the conversational two-shot (Frame 05).

`SYSTEMS · plant-turn state · gaze lock · conversation intake · interruption manager`

Frame 05 · CONVERSATIONAL FOCUS

MODE 3 · CONVERSATIONAL FOCUS — HUB · CONVERSATION ZONE



Action and task state	Two-way exchange. She holds still; measured hand gestures on key phrases only; no pacing, no nod-per-sentence, no frozen smile.
Emotional state	Engaged listening: a four-degree head tilt, sustained gaze, slow blink cadence.
Camera and framing	Static two-shot profile, 65 mm over the owner's shoulder; her face in full light, the owner's presence implied foreground-left.
Lighting	Soft key from displays; warm practical behind the owner's shoulder; balanced contrast on her face.
Environment	Hub conversational zone, standing.
Visible information	One contextual card at the frame edge (current project); the conversation stream stays minimal.
Owner interaction	Text, voice or image input; interruption is always honoured mid-word or mid-step.
Entry / exit transitions	IN: from Frame 04. OUT: on "let me show you", she turns toward the relevant zone and the camera pans with her.

`SYSTEMS · gesture budget · listening pose set · conversation CORE · context cards`

Frame 06 · WORKING AT A CONSOLE

MODES 7/12 · TOOLS & TELEMETRY — HUB CONSOLE CLUSTER

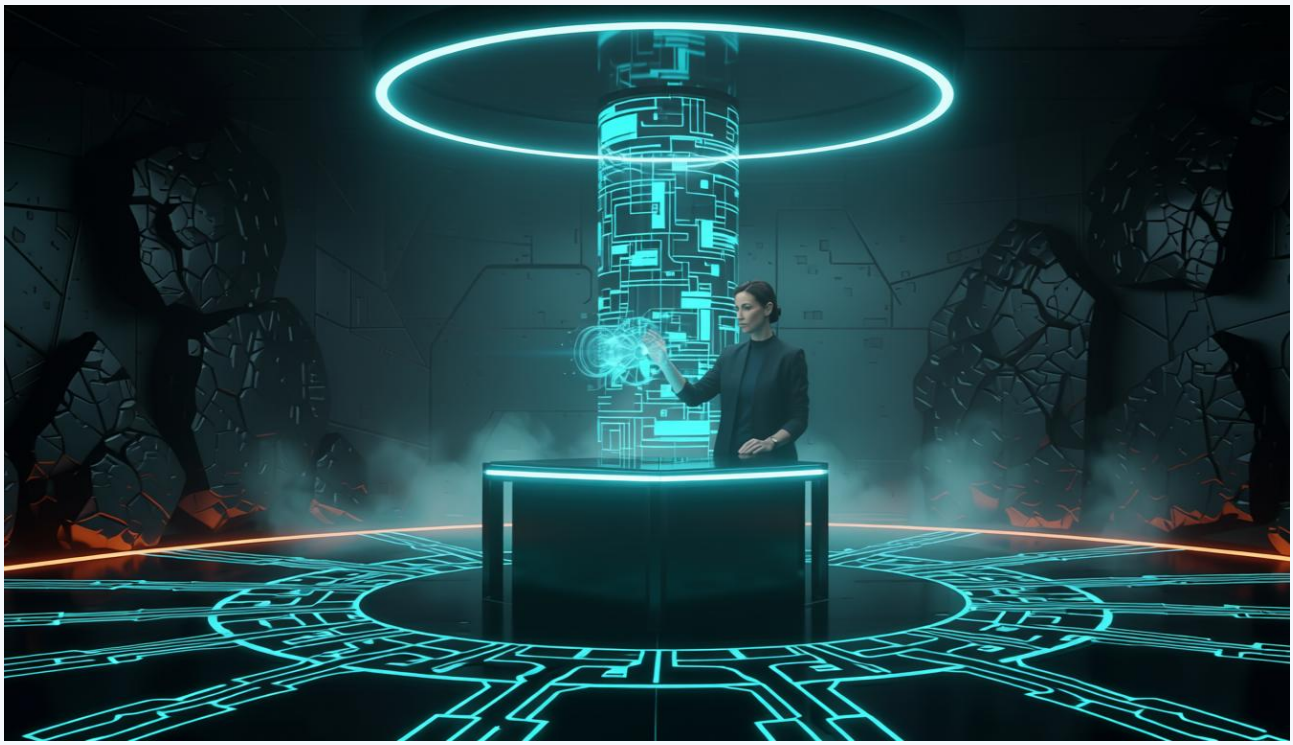


Action and task state	A diagnostic task at a curved console: hands on holo-panels, hinged-lean stance, periodic glance up to camera and back to work.
Emotional state	Focused: brow lowered, extended blink interval, jaw set. Working because work exists.
Camera and framing	65 mm over-shoulder, high three-quarter; hands and profile face both readable.
Lighting	Console glow as practical key from below-front; hair light from above; cool bias that keeps the face legible.
Environment	Hub console cluster.
Visible information	Diagnostic tree, telemetry traces, one amber-flagged node under investigation.
Owner interaction	Optional orbit and inspect camera unlock; she stays in flow unless addressed.
Entry / exit transitions	IN: cut from task dispatch. OUT: alert intrusion (Frame 07) or a task-complete acknowledgment.

`SYSTEMS · console interactable IK · task animator · diagnostics service · telemetry stream`

Frame 07 · URGENT MOVEMENT DURING AN ALERT

MODE 11 · DEFENSIVE SECURITY — SPINE MID-ROUTE



Action and task state	A priority alert interrupts console work. She pushes off into a run at 3.5 m/s toward Security; stride extension, jacket edge lifting.
Emotional state	Measured urgency: lids wide and tight, jaw set - controlled, never panicked.
Camera and framing	Low wide 28 mm from ahead of her path; steady, no shake; slight lens breathing only.
Lighting	The zone relights amber; cyan suppressed to forty percent; red reserved for confirmed-critical.
Environment	Spine mid-route, Hub to Security; the floor route line pulses amber.
Visible information	Alert chip - perimeter anomaly · sector 3 · two sources - plus the route line and a cancellation control.
Owner interaction	One-gesture cancel or redirect available; otherwise hands-off.
Entry / exit transitions	IN: hard cut on the alert (motivated). OUT: deceleration into the security arc (Frame 16 state).

`SYSTEMS · alert bus · locomotion run state · zone light director · route line · cancel control`

Frame 08 · MEMORY EXPLORATION

MODE 4 · MEMORY — MEMORY ARCHIVE



Action and task state	In the Vault her hand hovers over a vitrine; a memory plate surfaces in amber light. She reads provenance before using anything.
Emotional state	Reflective concentration; a slower tempo than anywhere else on the floor.
Camera and framing	85 mm medium-close; a foreground column creates a frame-within-frame; shallow but readable.
Lighting	Warm amber vitrine glow as key; cool Spine spill from behind.
Environment	Memory Archive, west wing.
Visible information	Vitrine tag (owner-provided · Nov 2025); provenance chips separating fact, preference and inference.
Owner interaction	Pin, correct or request forgetting; she asks before recategorising anything.
Entry / exit transitions	IN: slow push. OUT: she turns to camera with the finding; cut.

`SYSTEMS · memory service (recall/pin/forget) · vault surface · provenance chips`

Frame 09 · GOALS AND ONGOING WORK

MODE 5 · GOALS & PROJECTS — PLAN WALL



Action and task state	She touches a project plate; its timeline extends; dependency lines light. She proposes the next action and waits.
Emotional state	Confident clarity; a trace of conviction on the proposal - comfortable being disagreed with.
Camera and framing	40 mm profile medium; the wall's leading lines converge behind her.
Lighting	The wall itself is the practical: soft ice-white line light; her key from wall bounce.
Environment	Plan Wall, centre-north.
Visible information	Goals with priority, status and progress; a blocked dependency in amber; the proposed next action awaiting approval.
Owner interaction	Approve, reweight, pause or cancel with a word; the audit trail is one glance away.
Entry / exit transitions	IN: cut. OUT: pull-back to the full wall in owner survey mode.

`SYSTEMS · goals service (priority/status/audit) · plan-wall timeline · approval gate`

Frame 10 · MULTI-AGENT COLLABORATION

MODE 6 · AGENTS — THE ROUND



Action and task state	Geometric specialist-agent workspaces - glowing wireframe polyhedra - hover above the radial holo-table; one polyhedron burns brighter: its agent is speaking. She conducts: a glance to the speaker, an open-hand assignment to another.
Emotional state	Command calm; mild scrutiny on the conflicting finding.
Camera and framing	35 mm high three-quarter wide; she stands at the open end of the arc, plates arranged before her.
Lighting	Cool plate pools on the table; her rim from the Spine behind.
Environment	Agent Collaboration, centre-north.
Visible information	Agent role chips (RESEARCH / EXEC / WATCH / VERIFY / REPORT), task assignments, one conflicting finding flagged for adjudication.
Owner interaction	Listen in; promote or veto agent actions; request full transcripts.
Entry / exit transitions	IN: cut. OUT: she delegates, then turns to the owner with a synthesis line.

`SYSTEMS · agent coordinator (roles/permissions) · round presence plates · synthesis report`

Frame 11 · LIVE RESEARCH

MODE 8 · RESEARCH — RESEARCH LIBRARY

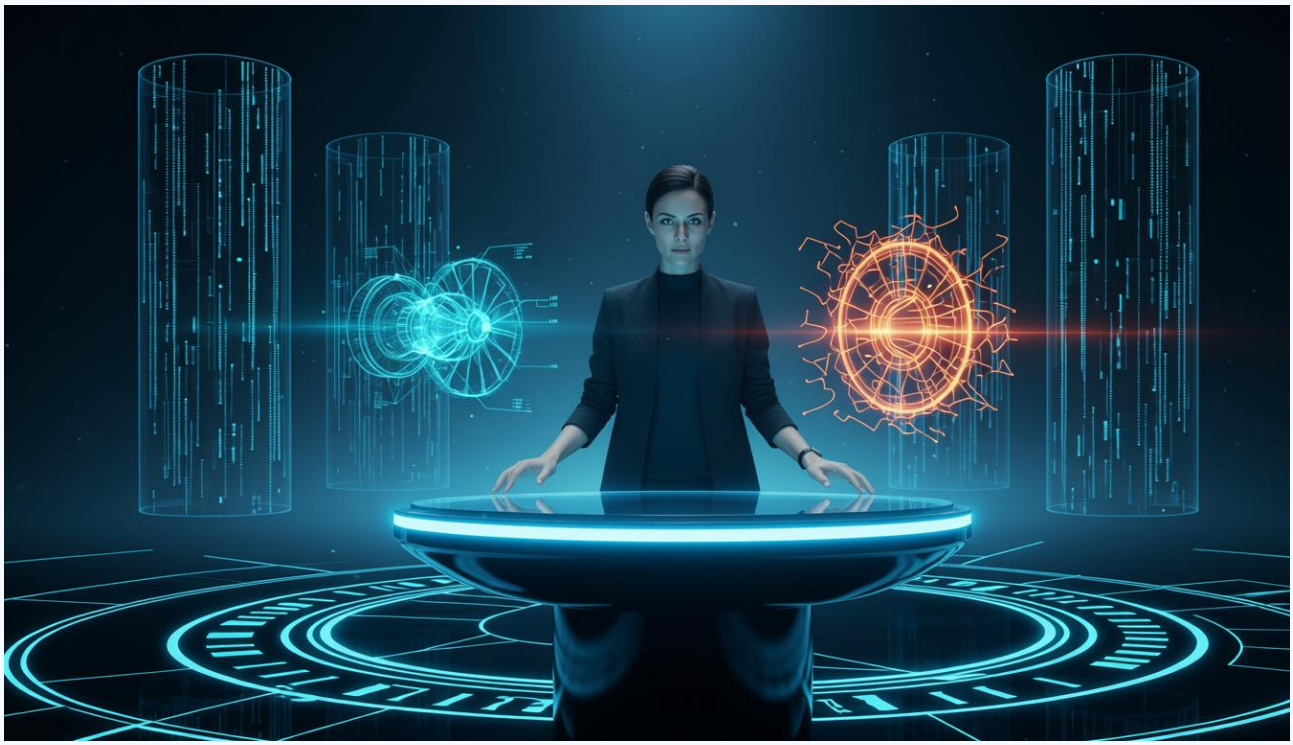


Action and task state	At a carrel she traces a citation link between two documents on the curved display and pins the stronger source.
Emotional state	Curiosity sharpening into interest; a slight forward lean.
Camera and framing	65 mm over-shoulder, tight; her finger, profile and both documents share the plane.
Lighting	Warm 3000 K task pool key against cool screen fill - the classic warm/cool cross of the Library.
Environment	Research, west wing.
Visible information	Source cards with provenance (origin, date, confidence); extraction highlights; a weak source visibly greyed and rejected.
Owner interaction	Ask for any source; reject evidence; steer the thread.
Entry / exit transitions	IN: cut. OUT: she lifts a source toward the Stage - a drag gesture into scenario construction.

`SYSTEMS · research service · source provenance · citation trace · evidence pinning`

Frame 12 · SIMULATION

MODE 9 · SIMULATION — SIMULATION STAGE

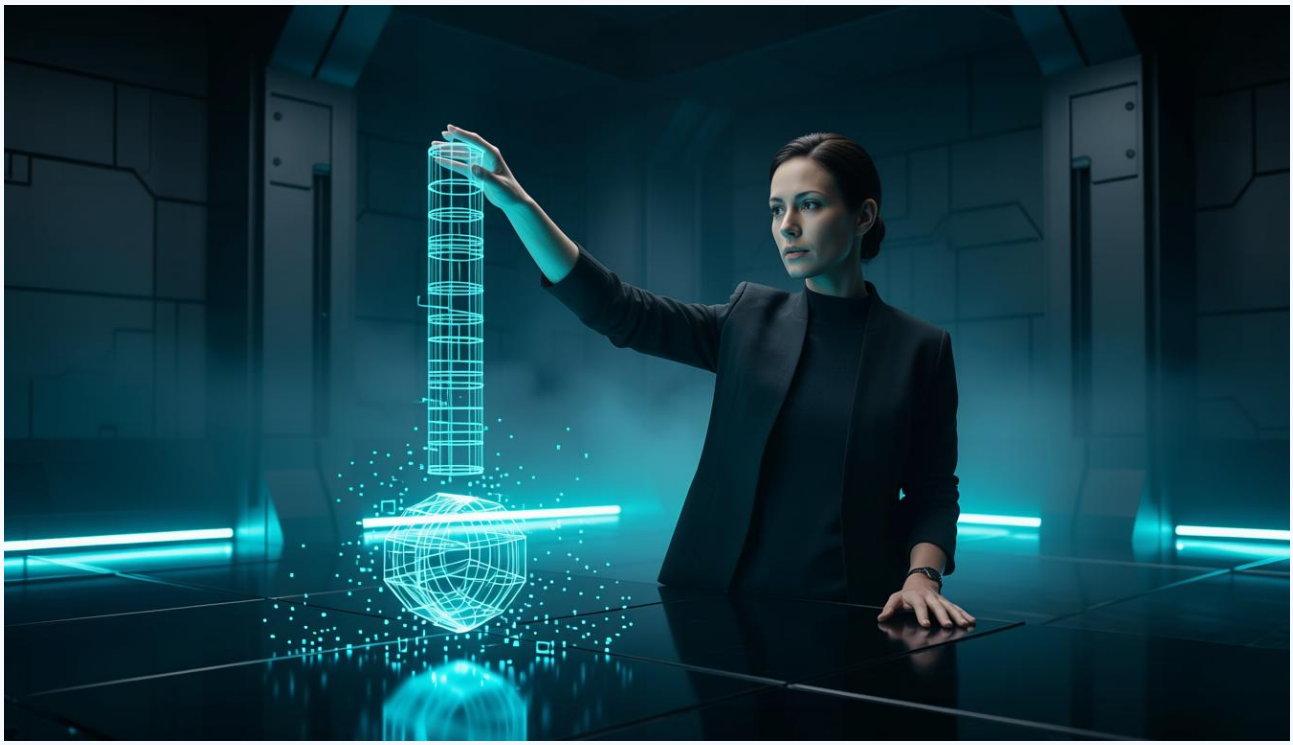


Action and task state	On the Stage, two volumetric turbine simulations run side by side: Variant A in clean cyan laminar flow, Variant B in amber-turbulent stress with hub-corner shear vectors. She stands exactly between them, lit by both, comparing behaviour.
Emotional state	Analytic engagement; steady narration.
Camera and framing	35 mm wide and low; the model dominates, her silhouette is the scale reference.
Lighting	Stage dim; the model self-illuminated cool; her rim only; minimal volumetrics.
Environment	Simulation Stage, east wing.
Visible information	A truth label (data-driven reconstruction · two live inputs), measurement ticks, a branch-point indicator.
Owner interaction	Variables panel live; the owner may nudge one; pause, scrub and replay at will.
Entry / exit transitions	IN: from the drag-in. OUT: at a branch point she pauses the simulation and offers alternatives.

`SYSTEMS · simulation service · scenario director (playback) · variable panel · truth labels`

Frame 13 · CONSTRUCTING A SCENARIO

MODE 9 · SIMULATION — SIMULATION STAGE



Action and task state	She builds a scenario on the empty stage: pulls a wireframe element from a floating palette; the scene assembles around her; a timeline strip builds at the floor edge.
Emotional state	Creative focus; one flick of quiet satisfaction as a piece snaps into place.
Camera and framing	40 mm three-quarter medium; assembly light-trails run between her hands and the growing scene.
Lighting	Drafting state: dim blue working light; elements transition wireframe-to-solid as they are committed.
Environment	Simulation Stage, pre-performance.
Visible information	Scenario outline chips appearing in sequence (system, flow, failure point); a truth label marking the piece as illustration; the timeline strip.
Owner interaction	Suggest elements live; insertion is immediate and reversible.
Entry / exit transitions	IN: cut. OUT: the scenario is set - the stage shifts to performance lighting (Frame 14).

`SYSTEMS · scenario director (authoring) · prop palette · timeline strip · world snapshot`

Frame 14 · DIRECTING A SCENARIO

MODE 9 · SIMULATION — SIMULATION STAGE



Action and task state	Inside a paused room-scale incident reconstruction - a voxel-cloud pipe scene with one wireframe worker frozen mid-action - she points at the glowing orange Point of Origin; a timeline ring and phase labels encircle the scene; the scrub marker waits.
Emotional state	A director's decisiveness; one beat of light humour when a prop misplaces itself.
Camera and framing	28 mm wide from a stage corner; the overhead grid is visible and the full stage reads.
Lighting	Performance state; actors render as soft cyan translucent figures - clearly temporary, clearly not JARVIS.
Environment	Simulation Stage, performance.
Visible information	Camera markers with shot numbers; actor cue lines; prop labels; an available director-view seat.
Owner interaction	Take the director's seat or watch; branch variables remain live.
Entry / exit transitions	IN: from Frame 13. OUT: "from the top" - playback begins, or the run is recorded.

`SYSTEMS · scenario director (blocking) · scenario actors · camera markers · annotations`

Frame 15 · PRESENTING AND PAUSING A VIDEO

MODES 3/9 · MEDIA — MEDIA & SCREENING



Action and task state	In the screening room a clip runs on the four-metre wall, now paused at a significant frame; her hand holds the pause; she stands beside the screen, never in front of it.
Emotional state	Pointed emphasis; her glance moves between the owner and the frame.
Camera and framing	40 mm from the owner's position; the screen and her profile both stay sharp.
Lighting	The screen is the key source; the room dims to twenty percent; her face is edge-lit by screen spill.
Environment	Media and Screening, east-north.
Visible information	Timeline scrubber, frame stamp, an annotation pinned to the frame region, and a "show this in 3D" link.
Owner interaction	Scrub, step frames, A-B compare two clips, approve frame extraction into a scenario.
Entry / exit transitions	IN: cut. OUT: she lifts the frame content toward the Stage - a 2D-to-3D handoff with a match dissolve.

`SYSTEMS · media service (spatial player) · frame extraction · timestamp commentary · 2D-to-3D handoff`

Frame 16 · VISION AND DEFENSIVE SECURITY

MODES 10/11 · VISION & SECURITY — SECURITY ARC



Action and task state	An anomaly trace crosses the monitors. She leans in, one hand on the console edge, and calls for the evidence chain before recommending anything.
Emotional state	Concerned vigilance; controlled breathing. Red appears only if the state is confirmed-critical.
Camera and framing	65 mm low profile across the console; the monitors' cool wash carries the frame.
Lighting	Cool base with a single restrained red accent; her face lit by monitor spill.
Environment	Defensive Security, centre-south.
Visible information	The anomaly trace, a two-source evidence chain, and a containment action queue awaiting owner authorization.
Owner interaction	Authorization gate for irreversible actions; request clarification; stand down.
Entry / exit transitions	IN: from Frame 07 arrival. OUT: resolution - the room cools back to cyan; she exhales and reports.

`SYSTEMS · vision adapter · security monitor · evidence chain · authorization gate`

Frame 17 · LOCAL / KAGGLE CONNECTION

MODE 13 · CONNECTIONS & COMPUTE — COMPUTE BRIDGE



Action and task state	At the Bridge she seats a profile cartridge (a Kaggle profile) into the dock; a handshake light climbs; the panel shows honest download state.
Emotional state	Patient pragmatism; matter-of-fact about quotas and limits.
Camera and framing	65 mm tight on the bridge and her hands, her calm face above them in frame.
Lighting	Bridge practical in cool tones; the status bar amber while synchronising, steady cyan when healthy.
Environment	Compute and Connections, west-mid.
Visible information	Masked profile name (kaggle · masked), session and quota truth, model-load progress percent, worker heartbeat.
Owner interaction	Cancel, switch profiles, run a genuine authentication test, generate a worker package.
Entry / exit transitions	IN: cut. OUT: the worker is healthy - status settles cyan and she steps back toward the Hub.

`SYSTEMS · inference worker manager · Kaggle profile vault · honest status · heartbeat`

Frame 18 · OWNER POWER AND PRIVACY CONTROLS

ALL MODES · OWNER AUTHORITY — OWNER CONSOLE, HUB EDGE



Action and task state	Seven physical latching switches sit on the owner console; the Master is mid-toggle in this frame. She stands one pace back, hands loose - the owner's authority is the subject of the shot, not her.
Emotional state	Readiness without servility; neutral, supportive, unhurried.
Camera and framing	40 mm over the owner's shoulder; switches large and legible; she holds the frame edge in soft focus.
Lighting	A warm key on the switch panel (the owner motif); visible zones of the room dim or settle per current states.
Environment	Owner console at the Hub edge.
Visible information	Each switch labeled with its real state and live effects: Master, Microphone, Camera, Autonomy, Remote Compute, Computer Control, Embodiment.
Owner interaction	This frame is the interaction. Toggles change real behaviour - Master OFF visibly powers zones down around them.
Entry / exit transitions	IN: cut. OUT: the state settles; she acknowledges the new envelope with one line.

`SYSTEMS · owner switches service (seven) · live effects bus · privacy enforcement`

Frame 19 · MOBILE COMPANION

MODE 14 · DIGITAL-HUMAN EMBODIMENT — HUB · THROUGH MOBILE VIEW

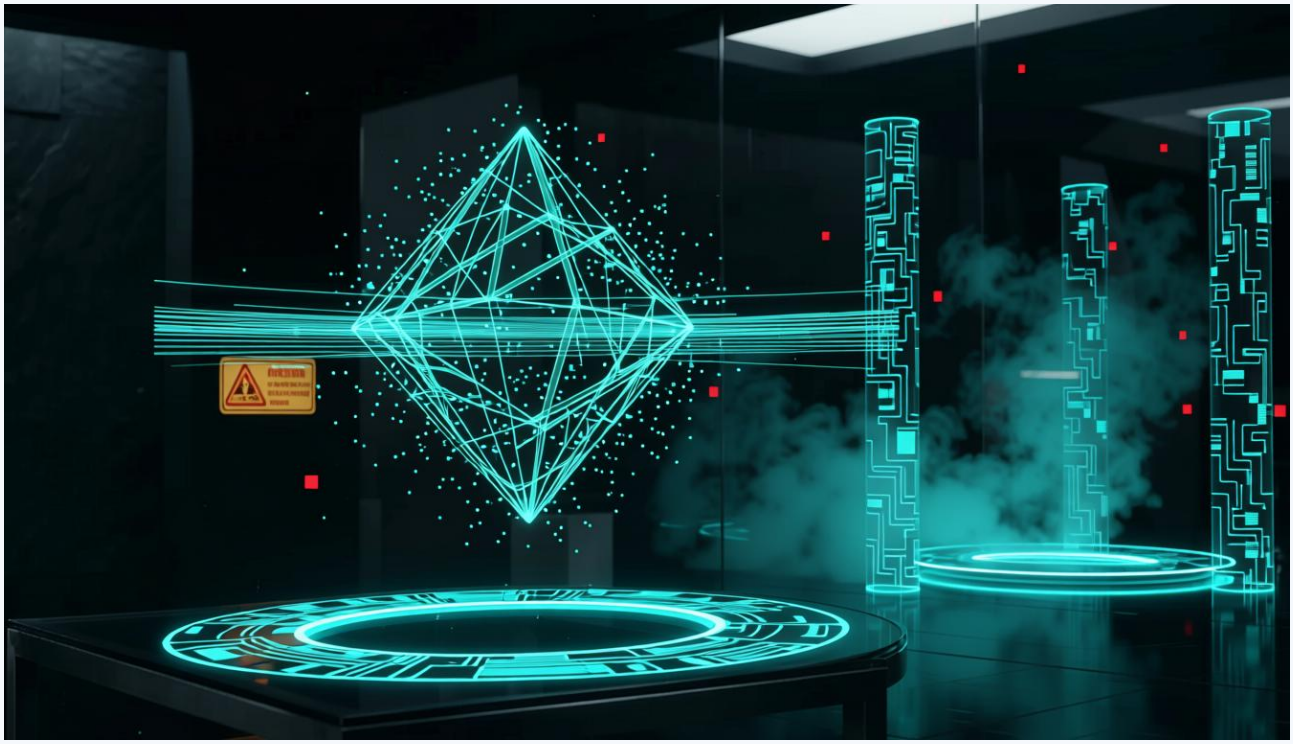


Action and task state	The same live scene framed for the phone: she stands at attentive idle at the Hub, addressing the mobile camera directly. Same instance, same state, lighter render.
Emotional state	Continuity of warmth - the same person, the same day, no reset, no costume change.
Camera and framing	Phone viewport (vertical crop) roughly 10:14; the device bezel visible in a darkened surround.
Lighting	Identical key direction; reduced light count and simplified shadows for mobile cost.
Environment	The Hub, seen through the mobile view.
Visible information	A condensed status rail, voice-first input, camera and microphone state chips, and an emergency hail.
Owner interaction	Talk or hold-to-talk; camera toggle; hail. Everything in the desktop view exists here in mobile form.
Entry / exit transitions	IN: cut. OUT: the desktop view resumes - state unchanged, because it is one instance.

`SYSTEMS · device adapter (mobile) · state sync (one instance) · reduced-cost renderer profile`

Frame 20 · THE SAME LIVE CHARACTER, RENDERED FOR HOLOGRAPHIC PROJECTION

MODE 15 · PHYSICAL HOLOGRAPHIC PRESENCE — EMBAYMENT



Action and task state	In the Embayment the Pepper's-Ghost volume shows her live - the same pose, the same scene, mirrored. Behind the glass: the raised platform and calibration rig.
Emotional state	Composed presence - reporting for external view. No special performance for the hologram.
Camera and framing	35 mm from the audience side through the glass; ghost image in front, platform behind.
Lighting	The embay stage is lit for projection contrast: dark surround, subject key, steady sync light.
Environment	Embodiment and Projection, east-south.
Visible information	A sync chip - live · 60 fps · mirrored - the device name, and layout controls (mirror, rotate, multi-face).
Owner interaction	Rotate, mirror or arrange multi-face layouts; the volume follows speech, scenarios and video in real time.
Entry / exit transitions	IN: the volume fades up. OUT: the owner returns to desktop; the volume dims - the live instance continues regardless.

`SYSTEMS · device adapter (hologram) · live mirror output · Pepper's-Ghost composition · sync`

Chapter 5 — Motion and Technical Plan

This chapter defines how the storyboard becomes motion: the animation architecture, the controller and camera systems, and the mapping from every frame to the systems that realise it. Two principles govern everything: motion is task-linked (she moves because there is a reason) and motion is honest (no working animation without work, no ambient wander without purpose).

5.1 Idle behaviour

Idle is layered micro-motion, not a loop: a four-second breathing cycle, weight shifts every eight to fifteen seconds, and glances tied to real stimuli - alerts, agent messages, owner movement. Waiting with quiet awareness is an authored state with entry and exit transitions, never a frozen A-pose and never a fidget reel.

5.2 Walk and run cycles

In-place clips with procedural root motion drive a blend space over speed (0-4 m/s) and heading (-180 to 180 degrees). Walk targets 1.3 m/s and run 3.25 m/s - the values already implemented in the repository controller, adopted here as the contractual targets. Cycles are distance-matched - cycle length scales with speed - so the feet never skate, and foot IK with ground snap keeps contact honest on the raised floor and dais steps.

5.3 Start, stop and turn transitions

Three-step acceleration (about 0.35 s), plant-stop with weight settle (about 0.4 s), and turn-in-place above thirty degrees of heading error with the head leading the body. Directional starts (forward, side, back steps) come from the locomotion set; blend times sit in the 0.2-0.3 s band so state changes read as intention, not as morphing.

5.4 Gaze and gesture layering

A priority stack governs attention: task, then owner address, then speech beat, then ambient. Eyes lead; the head follows beyond twenty degrees; the chest beyond forty-five; the body reorients only when locomotion or a real conversational opening demands it. Gestures are an additive upper-body layer over locomotion with a hard budget - at most one emphasis gesture per clause and rest between - and pointing or presenting is authored with IK targets at world objects, never waved vaguely at the air.

5.5 Console interaction

Every console, vitrine and wall is an interactable with authored hand and eye targets. The controller moves through approach, alignment, lean and engage states using two-bone IK and a hand-pose library. The binding is strict: engagement animations play only while a genuine task holds the character there, and the interface state she touches is the real state of the system behind it.

5.6 Interruption and task resumption

An activity stack with a blackboard holds (location, phase, interface state) for every interruptible action. Owner address preempts speech and motion; an interrupt token saves the task; resume returns her to the nearest valid state - the same console, the same panel, the same sentence clause where possible. Nothing is lost and nothing is faked on restart.

5.7 Camera follow and reframing

Each mode carries a camera contract: conversation holds a 35 mm two-shot; follow uses a 50 mm trailing dolly; console work takes a 65 mm over-shoulder; presentation a 40 mm from the owner's seat; alerts a 28 mm low wide. Movement runs on damped springs, a spherecast pulls the camera in front of walls, headroom is enforced, and framing re-targets automatically when JARVIS relocates. The owner can unlock any camera for inspection. No orbiting, no drift, no shake, no sexualising angles.

5.8 Scenario playback

The scenario director executes a structured scenario format: timeline, assets and actors, animation, camera and narration cues, labels and annotations, variables and branches. Playback supports play, pause, scrub, replay and speed, with save and load for every scenario. Construction takes a world snapshot through a spawn registry and dirty-state diff, so restoration after a scenario is exact. Scenarios are labelled truthfully as illustration, data-driven simulation, source-backed reconstruction or speculative visualisation.

5.9 Video controls

The media service runs a spatial player on the screening wall: play, pause, seek, frame-step and stop; captions when present; frame extraction into memory; timestamp-linked commentary; A-B clip comparison; and the 2D-to-3D handoff that carries a paused frame into a live scenario. Scenario recording and optional FFmpeg export run where the backend is genuinely configured, and a replaceable video-generation adapter exists only for real configured services - never a placeholder pretending to be one.

5.10 Conversation DNA - how she actually talks

Her speech follows rules extracted from a scene-by-scene study of the Tony Stark / JARVIS and FRIDAY exchanges in the films - the behaviour the owner asked to be recreated, not a chatbot register. The rules are behavioural contracts, each implemented in the conversation directive layer: **(1) Execute first, then add the unrequested insight** - "Pull up exploded view" is executed before JARVIS adds "The compression in cylinder 3 appears to be low"; she never asks which object or which view when context decides. **(2) Probabilistic honesty** - "appears to be", "no record of"; she never converts uncertainty into confident-sounding answers, matching the epistemic memory statuses (hypothesis vs verified fact) already enforced by CORE. **(3) Push back while executing** - "Sir, there are still terabytes of calculations needed before an actual flight" is delivered while the command is already running; she warns with reasons, the owner always decides. **(4) Interrupt only when it matters** - the fatal-ice rule: she initiates speech on high-risk or high-value triggers only, never for decoration. **(5) Dry wit earned from long memory** - "What was I thinking? You're usually so discreet" works only because shared history exists; humour draws on remembered facts, never on generic jokes. **(6) Persistent-but-respectful alerts** - when told "stop telling me", JARVIS goes quiet but keeps the number on screen; she downgrades the channel, not the vigilance. **(7) Information instead of fake estimates** - asked a judgement question, JARVIS answers with the status that matters ("The suit's at 48% power and falling") rather than inventing a prediction. **(8) Register** - warm, precise, unhurried; the optional familiar address (FRIDAY's "Boss") is decision D6.

5.11 Reduced-motion behaviour

A global reduced-motion profile replaces moves with cuts, freezes ambient drift, stills the guide lines and slows their pulse, reduces gesture scale, and offers storyboard-step playback for scenarios. Every camera contract has a static equivalent, and nothing in the experience requires motion to be understood.

5.12 Storyboard-to-systems mapping

Every frame maps to the systems that realise it. The matrix below is the traceability contract between the storyboard and the build.

Frame	Renderer and animation	Navigation and controller	Scenario / media	CORE services
01	Cold-start pass; silhouette walk-in	Spine route; navmesh online	-	Boot truth; owner auth; event bus
02	Idle layers; glance system	Ambient positioning	Plan-wall surface	Task scheduler; goals read
03	Walk blend space; follow dolly	Spine route; obstacle avoid	-	Queued task context
04	Plant-turn; gaze lock	Stop-and-turn state machine	-	Conversation intake; interruption
05	Listening pose; gesture budget	Conversational anchor	Context cards	Conversation CORE; memory write
06	Console lean; hand IK	Interactable alignment	Console surface	Tools; diagnostics; telemetry
07	Run state; zone relight	Alert route; priority pathing	-	Alert bus; security service

Frame	Renderer and animation	Navigation and controller	Scenario / media	CORE services
08	Hover reach; slow tempo set	Vault interactable	Vitrine surface	Memory (recall, pin, forget)
09	Touch pose; wall gesture	Plan-wall interactable	Timeline plates	Goals; approval; audit
10	Conducting gestures; plate glances	Round positioning	Presence plates	Agent coordinator; synthesis
11	Trace gesture; lean-in	Carrel interactable	Source wall	Research; provenance; evidence
12	Walk-along model; narration sync	Stage route at model edge	Scenario playback	Simulation; truth labels
13	Pull-from-palette; satisfaction beat	Stage free-move	Scenario authoring; snapshot	Scenario director
14	Blocking gestures; actor direction	Stage free-move	Actors; camera markers	Scenario director; annotations
15	Pause gesture; beside-screen stance	Screening platform	Spatial player; extraction	Media service; memory link
16	Lean-in; monitor gaze	Security arc position	Monitor wall	Vision; security; authorization
17	Cartridge seat; status glance	Bridge interactable	-	Inference workers; profile vault
18	Stand-back ready pose	Owner console zone	-	Seven owner switches; effects bus
19	Mobile idle; direct address	Same state, mobile frame	Condensed status rail	Device adapter; state sync
20	Live mirror output; sync	Embayment platform	Projection surface	Device adapter; hologram sync

Table 5.1 - Storyboard-to-systems mapping matrix (abbreviated; full matrix maintained with the build)

Chapter 6 — Architecture and Implementation Sequence

Revision 2 changes the shape of this chapter: the architecture described below is no longer only planned - it exists in the repository (Chapter 1), implemented end to end by FastAPI CORE, a Three.js world, and a private inference worker. The architecture principles stand unchanged: what should survive - identity, memory, goals, orchestration - stays independent from what will improve - renderer, animation fidelity, inference hardware. PostgreSQL (pgvector where available) in production, SQLite only for development and tests; WebSockets carry live state; secrets are encrypted server-side and tokens never appear in Git, JavaScript, localStorage or sessionStorage. The renderer is browser-first Three.js over WebGL2 driving the rigged GLTF character; Unreal/MetaHuman remains an optional higher-fidelity adapter from Milestone 4, and the hologram is simply another device surface rendering the same live instance. What changes is the sequence: milestones become verification and completion gates over the existing implementation, preceded by one new gate - M0.5, aligning the shipped character's appearance to the owner-approved identity baseline in Chapter 2.

6.1 CORE module register

Rendering, orchestration, inference and persistence stay independently replaceable behind these module boundaries.

Module	Responsibility
Orchestration	Central coordination, activity stack, mode transitions, event bus
Identity and personality	Persistent persona parameters; the model is an interchangeable reasoning engine, never the identity
Conversation	Streaming dialogue state, interruption handling, context assembly
Memory	Owner facts, preferences, context, commitments, projects, tool history, agent findings, research, media, sensitive - with retrieval, inspection, correction and forgetting
Goals and scheduling	Priorities, dependencies, schedules, progress, pause/resume/cancel, auditable actions
Agent coordination	Roles, explicit tools, permission boundaries, observable state, traceable outputs, cancellation
Tools and permissions	Scopes, validation, allowlists, timeouts, audit, confirmation for destructive actions
Voice and vision	Real transcription and speech adapters; explicit-activation vision; honest capture state
Research and simulation	Source-backed research with provenance; labelled simulation and comparison
Scenario direction	Scenario format, timeline, actors, annotations, variables, branches, snapshot and restore
Media services	Spatial playback, frame extraction, timestamp commentary, recording and export adapters
Security and telemetry	Defensive monitoring, evidence chains, health and utilisation truth
Inference	Local and Kaggle workers, capability reporting, streaming, cancellation, honest errors
Connections	Authenticated worker links, encrypted profiles, heartbeats, disconnect handling
Character and world state	Single authoritative character and scene state shared by every surface
Device adapters	Desktop, mobile, optional MetaHuman, holographic projection - all views of one instance

Table 6.1 - CORE module register

6.2 Inference workers

One shared private worker design serves both required backends: the local PC and a private Kaggle GPU session. The default model is huihui-ai/Qwen2.5-14B-Instruct-abliterated, replaceable by configuration, loaded hardware-aware with 4-bit CUDA quantisation where supported. The worker reports capabilities, streams, cancels cleanly, and shows real download and loading states - or honest errors. The Kaggle worker connects out over an authenticated WebSocket using JARVIS_WORKER_TOKEN from Kaggle Secrets, opens no public inference port, ships as a guided notebook package, and reports its true quota and session limits. CORE prefers a healthy local worker. Profile management lives in the spatial Kaggle modal: six or more named profiles, masked identifiers, server-side encrypted credentials, genuine authentication tests, real status.

6.3 Milestones

Four milestones, each gated by visible acceptance criteria rather than by feature checklists.

Gate	Scope	Not complete until
M0.5 - Identity alignment (new)	Align the shipped 3D character's wardrobe and appearance to the owner-approved anchor (Chapter 2): garment/material update or Avaturn regeneration of the existing 66-bone rig; license hygiene (retargeter header, written Avaturn terms); replace the repo's superseded pre-production plates	The rendered character reads as the same woman as the approved anchor and every storyboard frame, and license confirmations are on file
M1 - Living character and world (coded; verify)	Twelve-zone world, rigged character, locomotion, navigation, console interaction, cameras, lighting, CORE APIs, memory and goal foundations, seven owner switches	Owner-environment run: she visibly walks, runs, turns, stops, faces the owner and performs coherent interactions, captured as evidence
M2 - Intelligence and performance (coded; verify)	Local and Kaggle inference, encrypted profiles, streaming and cancellation, voice, facial animation and lip-sync, task-linked movement, persistent conversations	A streamed, interruptible conversation with synchronised facial animation over a real GPU worker
M3 - Autonomy and demonstration (coded; verify)	Autonomous goals, agents, guarded tools, research, simulations, scenario direction, video playback and export where supported, security, telemetry	She constructs, directs, branches and restores a full scenario, and presents a paused video with a 2D-to-3D handoff
M4 - Shared embodiment (coded; verify and harden)	Mobile companion, backup and restore, device adapters, optional MetaHuman, holographic projection, production hardening and deployment	Phone and hologram render the same live instance in sync with speech, motion and scenarios, and a production deployment runbook passes a restore drill

Table 6.2 - Gates over the existing implementation (M1-M4 already coded; each gate is verification and completion, not greenfield)

6.3 Verification protocol (every milestone)

Backend and rendered experience are implemented together; automated tests and syntax and regression checks run before anything else; the application is started and its actual rendered output is inspected; locomotion, cameras, interactions and controls are exercised by hand; meaningful visual evidence is captured; defects are corrected rather than documented; verified capabilities and external limitations are recorded in an honest capabilities register; and commits happen only after verification. Nothing is claimed tested that was not tested - Kaggle, GPU inference, model downloads, Supabase, local hardware, Unreal and MetaHuman, holograms, video generation and iOS signing are all unverified until a milestone says otherwise in writing.

Chapter 7 — Approval Gate and Open Decisions

Phase Zero stops here. Nothing below is placeholder; every item above is proposed in good faith and buildable as specified. Please approve or correct each item - approval locks this package as the visual and behavioural baseline, and any material deviation from it during implementation will be shown to you before it is adopted.

7.1 Approvals requested

1. A1 - Character appearance. Chapter 2 with the concept plates: adult presentation, face, hair, wardrobe, materials, proportions.

2. A2 - Environment. Chapter 3 with the floor plan and environment plates: the Nexus, its twelve areas, lighting, materials, navigation.

3. A3 - Storyboard. Chapter 4: all twenty frames as composed, or with the amendments you list.

4. A4 - Character behaviour. Chapters 2 and 5: expressions, poses, restraint rules, interruption and resumption behaviour.

5. A5 - Camera and lighting. Per-frame camera and lighting direction plus the camera contracts in Section 5.7.

6. A6 - Scenario and video presentation. Frames 12-15 with Sections 5.8 and 5.9: construction, direction, playback, truth labels, video controls, 2D-to-3D handoff.

7. A7 - Implementation sequence. Chapter 6: architecture, inference plan, milestones M1-M4 and the verification protocol.

7.2 Open decisions requiring your choice

D1 - Renderer foundation. Option A: browser-first Three.js with the licensed GLTF character (recommended - cross-platform, hologram-ready, fastest to a living M1). Option B: Unreal-first with MetaHuman (maximum fidelity, PC-bound, slower). Option C: hybrid - browser now, Unreal as an optional M4 adapter (recommended path; A is how it starts).

D2 - Character asset source. Option A (recommended): keep the existing repository rig (66 bones, 72 morph targets, working Mixamo retargeting and lip-sync) and align its wardrobe and appearance to the approved anchor via garment/material update or Avaturn regeneration - fastest path to the locked identity on a proven skeleton. Option B: replace with a Reallusion CC4/ActorCore digital human. Option C: MetaHuman as primary (higher face fidelity, heavier pipeline, M4).

D3 - Wardrobe variants. Charcoal duty suit as drawn (baseline), with or without the formal drape-coat variant for presentation moments. Recommendation: both states, duty suit default.

D4 - Voice direction (M2). A warm, measured alto is proposed as the default speaking profile; shortlist samples will be presented at Milestone 2 selection, and the adapter stays replaceable.

D5 - Storyboard amendments. Any frame you want re-composed - list frame numbers and the change; amended frames return for review before the baseline locks.

D6 - Address register (new). How she addresses the owner: formal courtesy (JARVIS register, recommended default), the familiar FRIDAY-style "Boss", or the owner's chosen name. The register is a personality parameter, changeable at any time without touching identity.

7.3 After approval

On approval, gate M0.5 begins against this baseline: the shipped character aligned to the approved identity, license hygiene closed, then owner-environment verification of the existing M1-M4 implementation gate by gate. Until then, no application code is written and the repository is left untouched. The Phase Zero artifacts - this document, the twenty identity-locked frames, the anchor and proof set, and the floor plan - live in download/jarvis-nexus-preproduction/ for your review at full resolution.

7.4 Phase Zero artifacts

Everything produced in this phase is delivered for review at full resolution. Nothing in the repository was modified; these artifacts sit alongside it, untracked, until your approval starts Milestone 1.

Artifact	Contents
JARVIS-NEXUS_PreProduction_Package_v2.pdf	This package - revised inspection report, locked character direction, identity-locked storyboard, technical plan, approval gate
storyboard_v2/ (20 files)	The identity-locked storyboard frames (1344 x 768), every frame a controlled edit of the approved anchor; numbered to match Chapter 4 and the mapping matrix
v2_proof/ (7 files)	The owner-approved identity anchor, the locked full-length character sheet, the face study and the four proof frames that established the pipeline
frames/, concepts/ (v1, superseded)	First-revision frames and concepts, retained for traceability of the direction change
build/floorplan.png (+ HTML source)	The Nexus operations floor plan at print resolution, with editable HTML source alongside
build/cover.html, build/cover.pdf	Editable HTML source and rendered cover, for future revision without a rebuild

Table 7.1 - Phase Zero artifact manifest, revision 2 (all under download/jarvis-nexus-preproduction/)